

Lab 3

IS

TU257

Lab 3 solution

Business Requirements

A company has several departments. Each department has a supervisor and at least one employee. Employees must be assigned to at least one, but possibly more departments. At least one employee is assigned to a project, but an employee may be on holidays and not assigned to any projects.

Work flow

Step 1: Identify Entities

Step 2: Find Relationships

Relationship matrix					

Step 3: Draw Rough ER Diagram

Work flow

Step 4: Fill in Cardinality

Step 5: Define Primary Keys

Step 6: Draw Key-Based ERD (eliminating many to many relationships)

Step 7: Identify Attributes & Step 8: Map Attributes

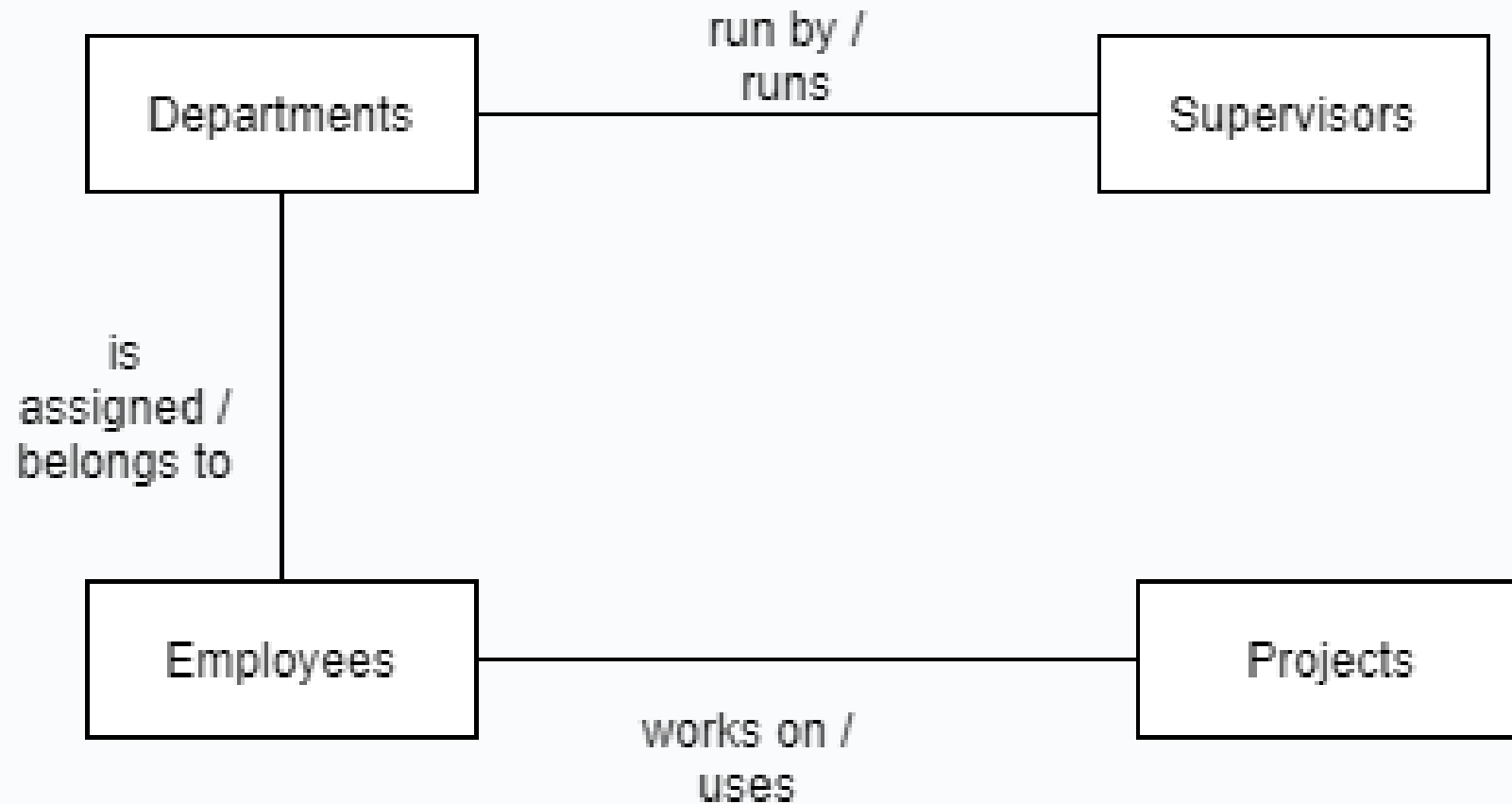
Step 9: Draw Fully Attributed E-R Diagram

Lab 3

- Entities: Employees, Supervisors, Departments, Projects
- Relationships:

	Departments	Employees	Supervisors	Projects
Departments	x	-----	-----	-----
Employees	belongs to/is assigned	X	-----	-----
Supervisors	Run/run by		x	-----
Projects		Uses/works on		x

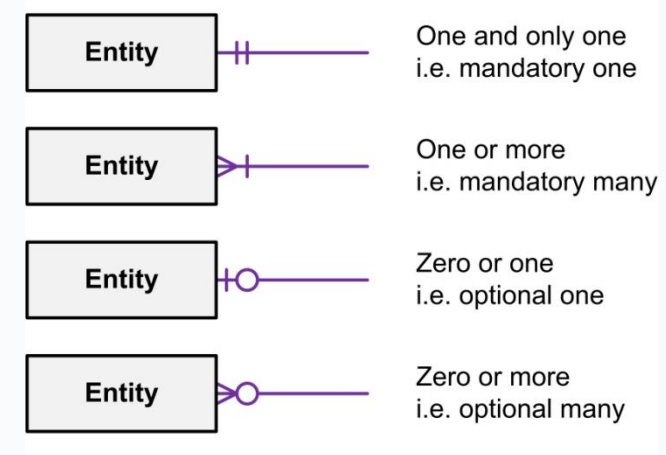
Lab 3 Rough ER Diagram



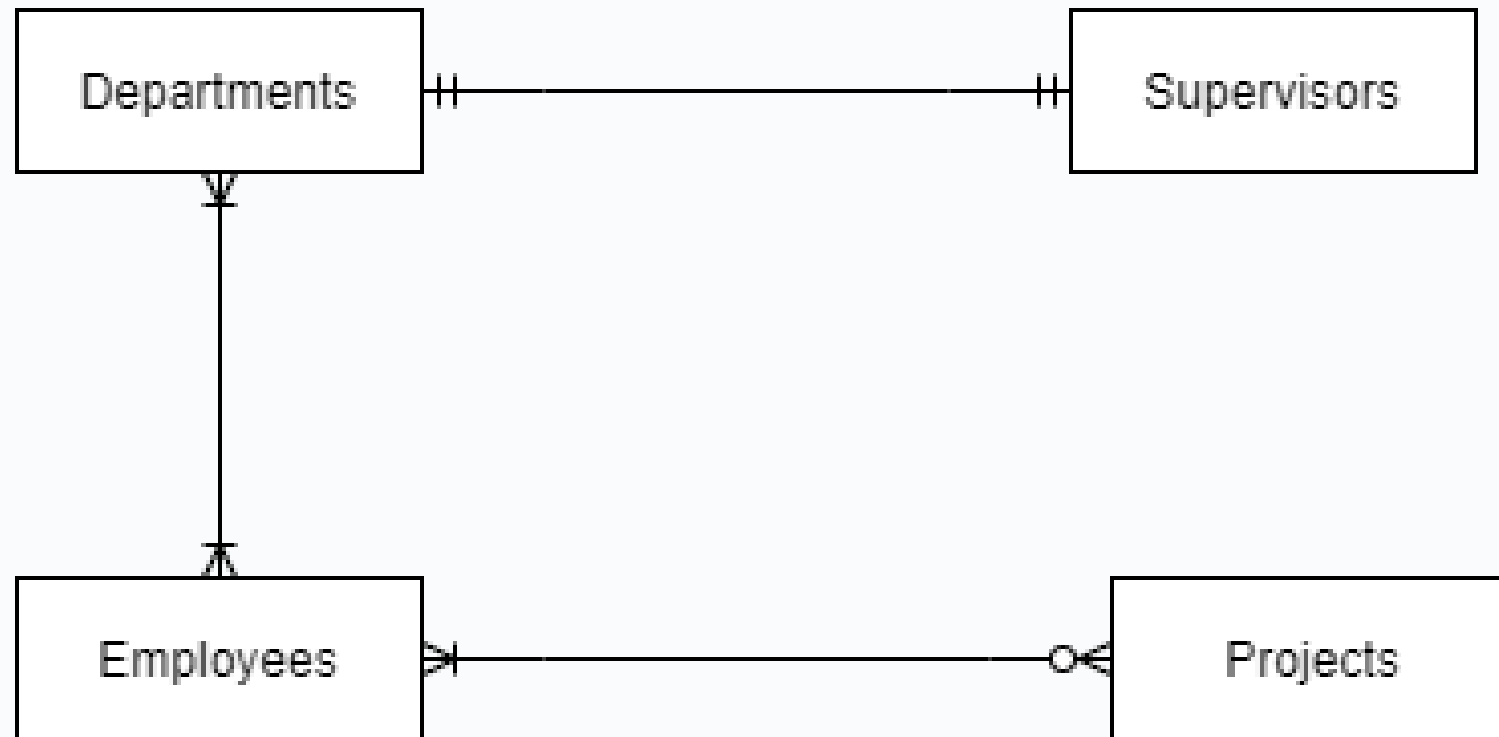
Lab 3

- Cardinality:

- Each **department** has exactly one **supervisor**.
- A **supervisor** is in charge of one and only one **department**.
- Each **department** is assigned at least one **employee**.
- Each **employee** works for at least one **department**.
- Each **project** has at least one **employee** working on it.
- An **employee** is assigned to 0 or more **projects**.

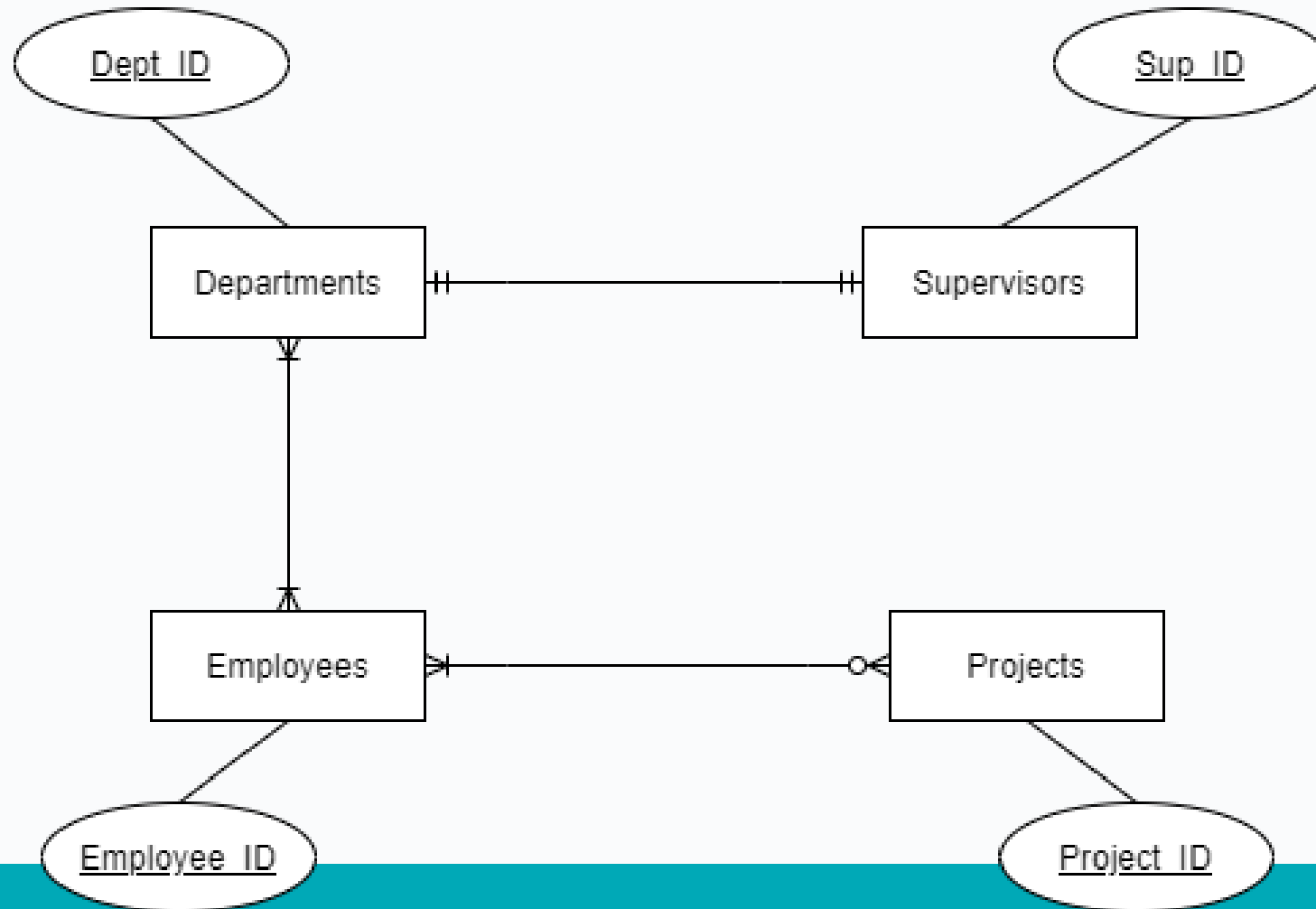


Lab 3 Cardinality

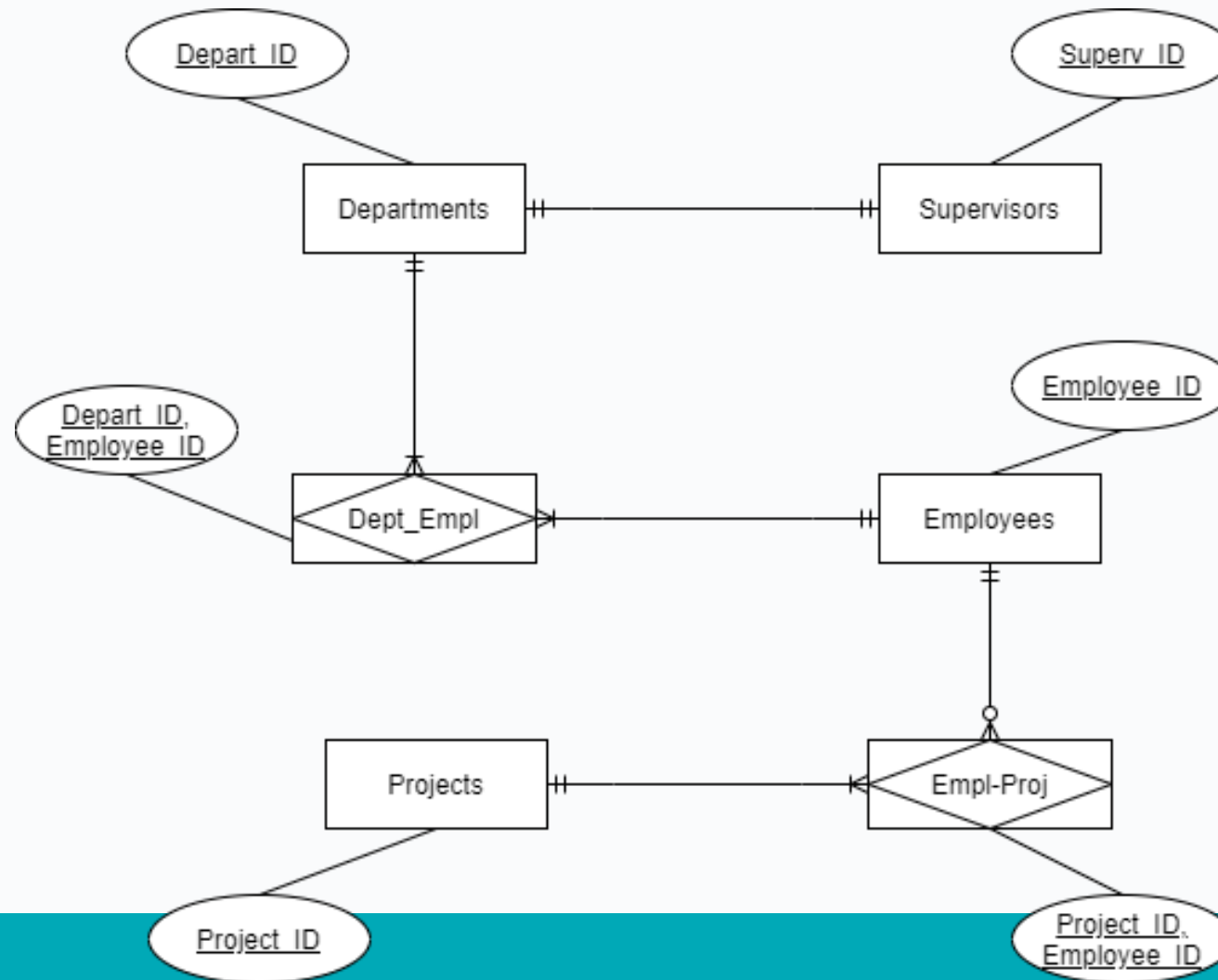


Primary Keys

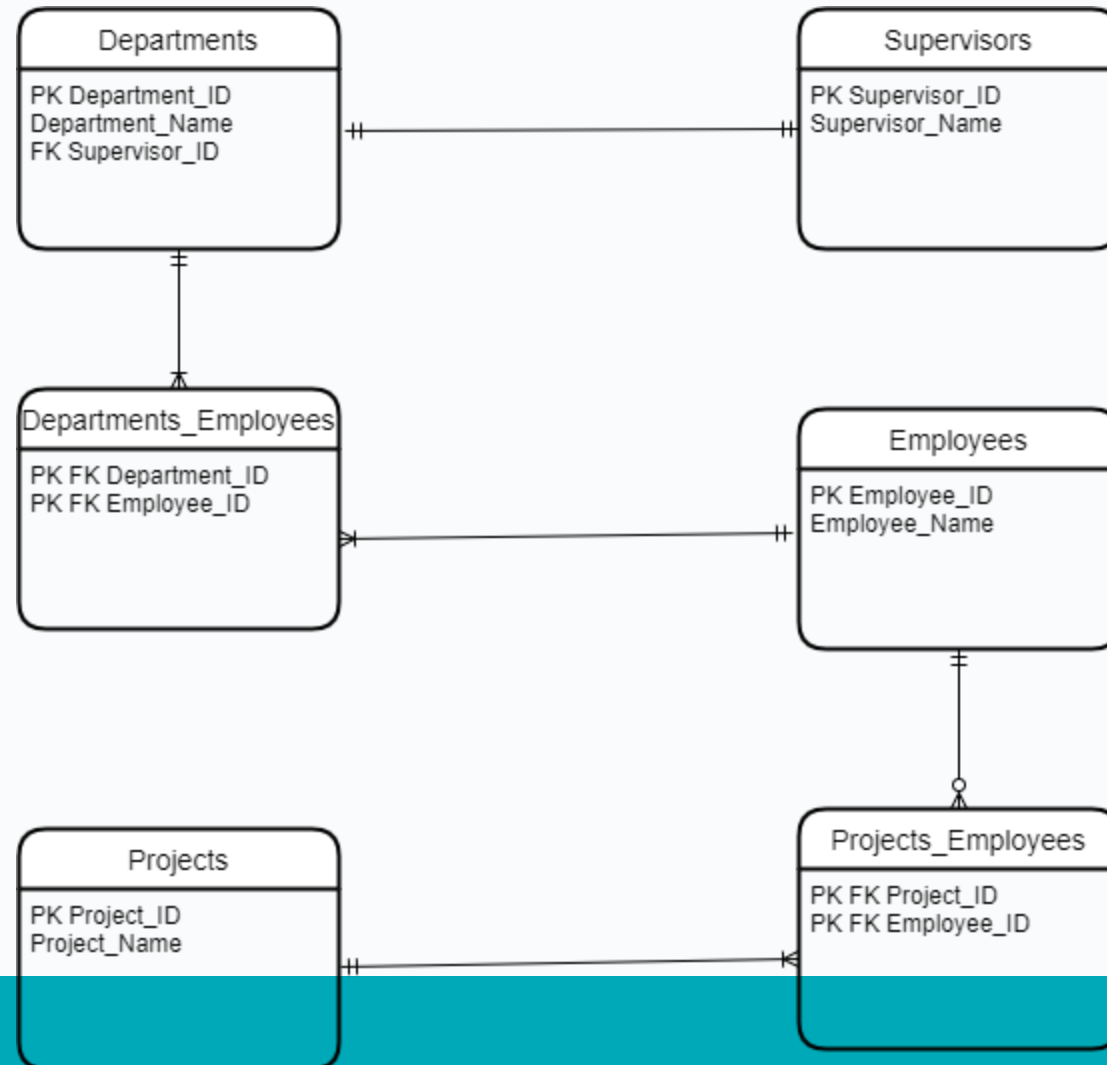
Rough ER Diagram *



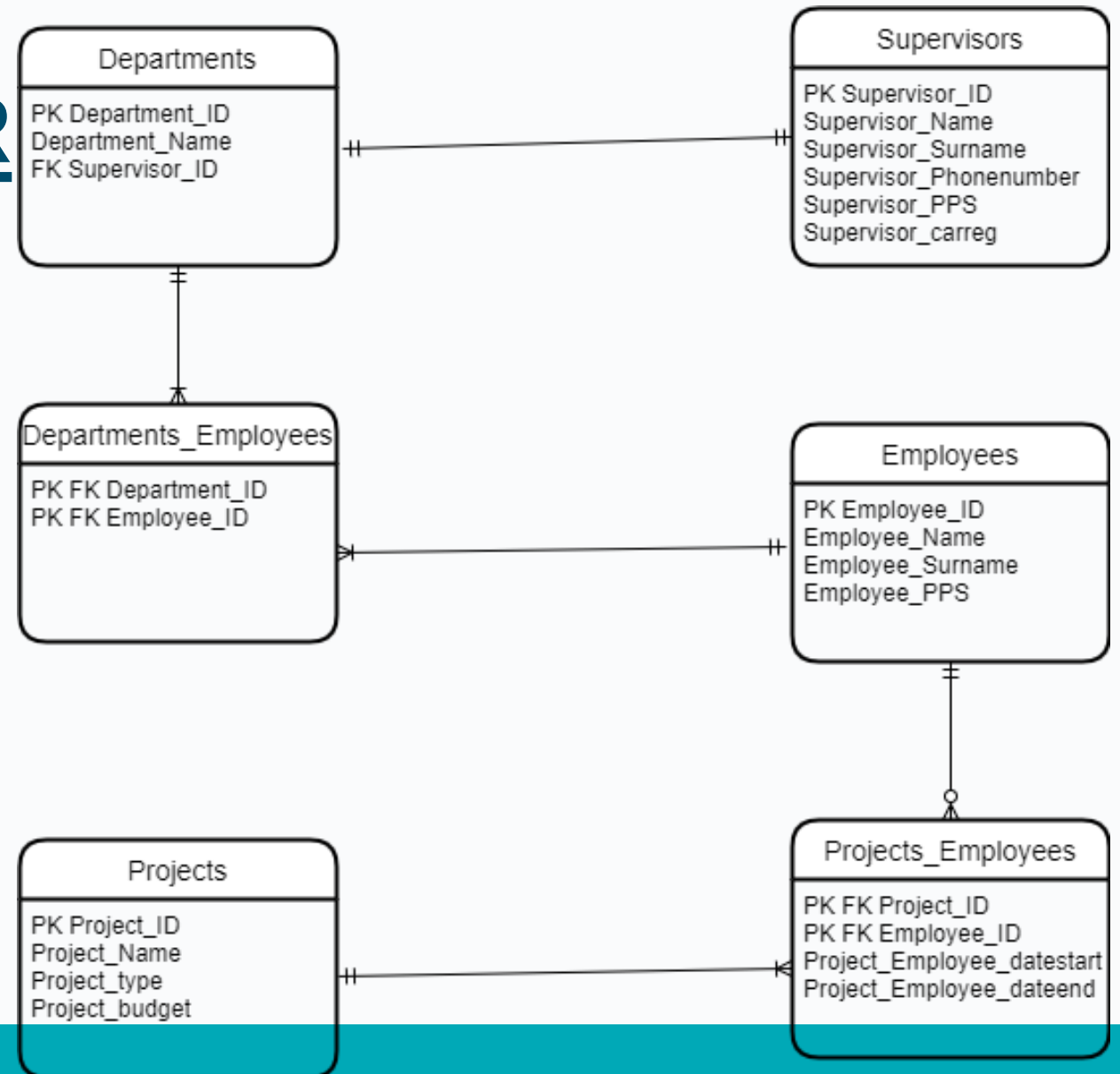
Remove Many-to-Many Relationships



Key Based ER diagram



Identify Attributes & Map Attributes Fully Attributed E-R Diagram



Version 1 Assumptions

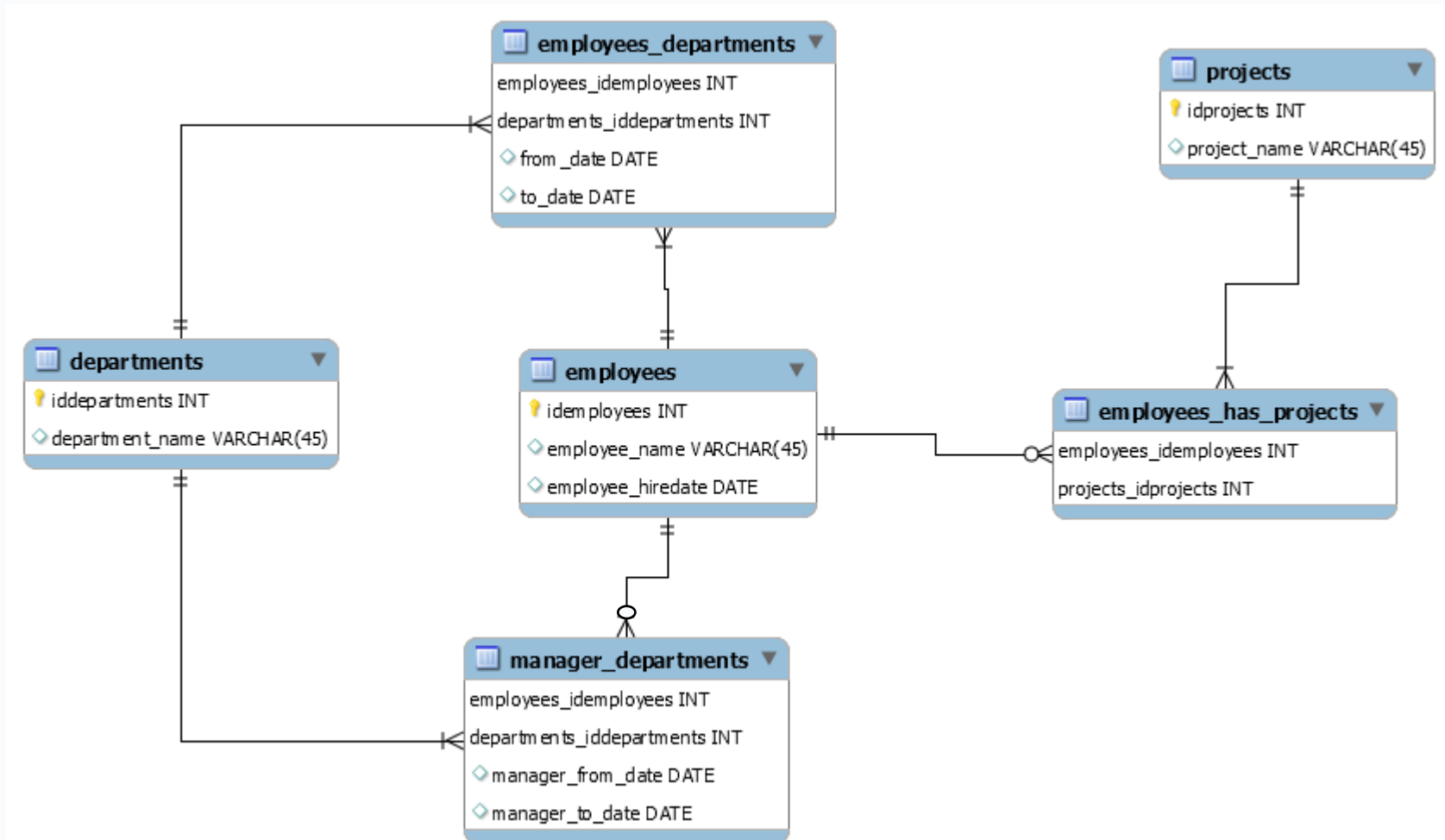
1. Staff is hired into a role, either employee or supervisor
2. Supervisors are given a company phone and a company car
3. Supervisors deal with the management of departments but do not work in projects

Version 1 Assumptions

- 4. Supervisors do not get assigned into projects
- 5. Employees can be assigned to different projects
- 6. Employees are not assigned to the same project for different periods of time

Lab 3 Version 2

- Supervisor as employee



Lab 3 Version 2 Assumptions

1. Any employee can be assigned the role of supervisor for a period of time
2. Supervisors don't have any special attributes (compared to a normal employee)
3. Supervisors may work on projects

Lab 3 Version 2 Assumptions

- 4. Employees can be assigned to different projects
- 5. Employees are not assigned to the same project for different periods of time

Version 3 Student contribution

- Discussion forum
 - What if employees can only work in one project at a time?
Do we want to keep a record of what projects they worked on or not?
 - What if employees could be assigned to the same project at different periods of time?